JEFFREY (JEB) SONG

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Department of Physics, University of Washington

EDUCATION

University of Washington (UW) *2023-Present*
B.Sc. in Physics (Honours), B.Sc. in Mathematics, 3.94/4.00.
CSE 599: ML for Neuroscience (4.0), CSE 373: Data Structures and Algorithms (4.0), CSE 374: Intermediate Programming Concepts and Tools (4.0), MATH 424-426: Fundamental Concepts of Analysis Series (4.0), PHYS 324-325: Quantum Mechanics I/II (4.0), PHYS 328: Statistical Physics (4.0), PHYS 521: Quantum Information (4.0)

National University of Singapore (NUS) *2021-2023*
B.Sc. in Physics, 4.36/5.00. *NUS College Programme (Honours College).*
IT1244: Aritifical Intelligence (A), CS1010E: Programming Methodology (A-), PC2130: Quantum Mechanics I (A+), UNL2210: Mathematics and Reality (A+)

PUBLICATIONS

J. Z. Song, H. Ha, W. W. Ho, and V. B. Bulchandani, *A theory of quasiballistic spin transport*, arXiv:2503.15756, 2025. `https://arxiv.org/abs/2503.15756`

RESEARCH EXPERIENCE

Rice University Asst. Prof. Vir Bulchandani Group —— *Quasi-Ballistic Spin Transport*
Sep 2024 - Present
Working on analyzing regimes for long-lived ballistic modes of spin. We are on track to submission to PRB in April.

NUS Asst. Prof. Ho Wenwei Group —— *1D Measurement-Based Quantum Computation*
Jan 2023 - Present
Studied 1D Measurement-Based Quantum Computation (MBQC) using tensor networks for my UROPS project. Reproduced the results of a novel 1D MBQC scheme using Python and extended their findings to accommodate higher logical qubit counts and qutrits. My UROPS report was nominated for the best UROPS award by the Physics Department.

UW Assoc. Prof. Samu Taulu Group —— *EEG Calibration via Quasi-Static Models*
June 2024 - Jan 2025
Developed and optimised EEG calibration algorithms informed by electromagnetic models, comparing both local and global approaches in `FieldTrip`. Simulated and analysed source localisation techniques.

UW Prof. Arka Majumdar Group —— *Optical Machine Learning*
Aug 2023 - Feb 2025
Exploring multi-input reservoir computing (a form of recurrent neural networks) using optical spiral waveguides. Developed simulation scripts in Lumerical (`varFDTD` and `FDE`) to design geometries and process results. Working on simulating input data for the etched device.

NUS Assoc. Prof. Kuldip Singh Group —— *Foundations of Quantum Mechanics*
Jan 2023 - May 2023
Explored ψ-ontological models, Bell's inequality, and the Pusey-Barrett-Rudolph theorem for directed reading project. Completed a project paper and presentation on limitations of such models.

NUS Asst. Prof. Lee Chinghua Group ——— *Non-Hermitian Quantum Physics with Qiskit*
May 2022 - Feb 2023
Simulated quantum systems using IBM's `Qiskit` package. Reproduced the SSH model, implemented trotterization for arbitrary unitaries, and decomposed non-Hermitian Hamiltonians.

NUS CQT Asst. Prof. Loh Huanqian Lab —— *Lasers for Neutral Atom Quantum Computing*
Feb 2022 - June 2022
Built an intensity servo system and constructed optical setups involving AOMs and DDS for precise control. Presented an internal talk on error correction techniques for laser servos.

ACTIVITIES

UW Undergraduate Physics Mentoring Program *Jan 2024 - Present*
Mentor
Provided academic and personal support to three mentees, facilitating bi-weekly meetings and participating in academic and social events.

UW Society for Physics Students (SPS) *September 2023 - Present*
Member
Participated in weekly talks by guest professors, attended various seminars and events hosted by UW SPS. Hosted an information session for pre-majors on physics.

NUSC Philosophy Club *June 2022 - June 2023*
Founder
I organised bi-weekly philosophy discussions, with a focus on the philosophical aspects of science and math. I successfully organized several talks and lectures by philosophy professors, providing an opportunity for students to learn from experts in the field and expand their understanding of philosophy.

NUS Physics Society *August 2021 - August 2022*
Events Director
I helped organise a successful physics mentor-mentee program. Furthermore, I was instrumental in jointly planning and organizing online meetups for physics students, providing opportunities for networking and learning beyond the classroom.

NUSC Basketball Club *August 2021 - August 2022*
Member
Participated in training sessions every week. Participated in the Inter-collegiate games basketball competition.

AWARDS

Mary Gates Research Scholarship *2024*

Guy Pitzel Scholarship (Departmental Honours) *2024*

Goldwater Scholar UW Nominee *2024*

CRISP Award for Best UROPS Nominee *2023*

TECHNICAL SKILLS

Technical Skills Python (Proficient), Linux (Ubuntu/Fedora), Java, PyTorch, Lumerical, Qiskit, Matlab, Bash/Zsh Scripting, C